Candidate Name	Centre Number	Candidate Number

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

214436

General Certificate of Education Ordinary Level

MATHEMATICS — 4008/1, 4028/1

PAPER 1

Friday **12 NOVEMBER 2004** Morning 2 hours 30 minutes

Candidates answer on the question paper.
Additional materials:
Geometrical instruments

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES
Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.
If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

Mathematical tables, slide rules and calculators should not be used.

INFORMATION FOR CANDIDATES
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE

This question paper consists of 24 printed pages.

[Turn over

For Examiner's Use

NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR CALCULATORS MAY BE USED IN THIS PAPER

1 **(a)** Express 9,846 correct to

(i) two decimal places,

(ii) one significant figure.

(b) Find the approximate value of $\sqrt{897,5}$, giving your answer correct to the nearest whole number.

Answer *(a)* *(i)* ______________________ [1]

(ii) ______________________ [1]

(b) ______________________ [1]

2 Evaluate

(a) $3 + 8\frac{1}{10} - 6\frac{4}{5}$,

(b) $2 \div 4 \times 3$,

(c) 0,55 m − 27 mm giving your answer in millimetres.

Answer *(a)* ______________________ [1]

(b) ______________________ [1]

(c) ______________________ mm [1]

For Examiner's Use

3 **(a)** Simplify

(i) $\left(3x^2\right)^2$,

(ii) $\dfrac{4-y}{8-2y}$.

(b) Write down the coordinates of the point where the line $y = 2x + 3$ cuts the y–axis.

Answer *(a)* *(i)* ______________________ [1]

(ii) ______________________ [1]

(b) (;) [1]

4 Evaluate

(a) -31^0,

(b) $64^{\frac{2}{3}}$,

(c) $\left(\dfrac{1}{5}\right)^{-3}$.

Answer *(a)* ______________________ [1]

(b) ______________________ [1]

(c) ______________________ [1]

For Examiner's Use

5 A polygon has 8 equal sides.

(a) State the special name of the polygon.

(b) Calculate the size of the interior angles of the polygon.

Answer *(a)* ____________________ [1]

(b) ____________________ [2]

6 Factorise completely

(a) $100x^2 - y^2$,

(b) $3m^2 - 7m - 6$.

Answer *(a)* ____________________ [1]

(b) ____________________ [2]

For Examiner's Use

7 (a) Write down \$645 correct to the nearest hundred dollars.

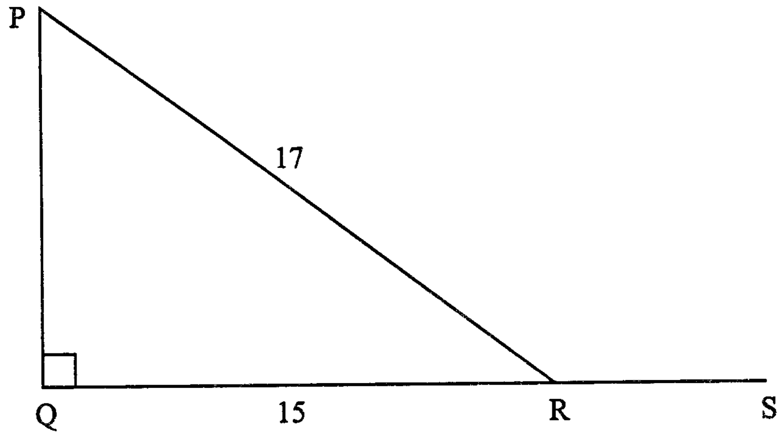
(b) Simplify $\frac{24}{600\ 000}$, giving your answer in standard form.

Answer (a) .... [1]

(b) .... [2]

For Examiner's Use

8



In the diagram, $P\hat{Q}R = 90^\circ$, PR = 17 cm, QR = 15 cm and QRS is a straight line.

(a) Calculate PQ.

(b) Find, giving the answer as a common fraction,

(i) $\cos P\hat{R}Q$,

(ii) $\tan P\hat{R}S$.

Answer *(a)* PQ = ____________ cm[1]

(b) (i) $\cos P\hat{R}Q$ = ____________ [1]

(ii) $\tan P\hat{R}S$ = ____________ [1]

For Examiner's Use

9 **(a)** Given that $\begin{pmatrix} 2 & x \\ 4 & 8 \end{pmatrix} - \begin{pmatrix} 1 & 3 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$, find the value of x.

(b) Evaluate $\begin{pmatrix} 2 \\ 3 \end{pmatrix}(4)$.

(c) State the name given to a matrix whose determinant is zero.

Answer *(a)* x = ______________________ [1]

(b) ______________________ [1]

(c) ______________________ [1]

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10 It is given that

$\xi = \{x\!: 1 \leq x < 15, x \text{ is an integer}\}$,

$A = \{x\!: x \text{ is a perfect square}\}$ and

$B = \{x\!: x \text{ is a multiple of } 3\}$.

(a) List the elements of

(i) A,

(ii) $A \cup B$.

(b) Write down the value of $n(A \cap B)'$.

Answer *(a)* *(i)* ______________________ [1]

(ii) ______________________ [1]

(b) ______________________ [1]

[Turn over

For Examiner's Use

11 Solve the simultaneous equations

$$4x - 7y = 23,$$

$$3x + y = -1.5.$$

Answer $x =$ ____________________

$y =$ ____________________ [3]

12 Evaluate

(a) $2130_5 + 4423_5$, giving your answer in base five,

(b) $1114_8 - 11110_2$, giving your answer in base ten.

Answer *(a)* ____________________ [1]

(b) ____________________ [2]

For Examiner's Use

13 Two solid spheres A and B are made of the same material. The radius of B is three times the radius of A and the surface area of A is 20 cm^2.

(a) Calculate the surface area of B.

(b) Given that the mass of A is 140 g, calculate, in kilogrammes, the mass of B.

Answer *(a)* ____________________ cm^2 [1]

(b) ____________________ kg [2]

14 (a) Express 00 13 as time in twelve-hour notation.

(b) A train left Harare at 20 30 and arrived in Bulawayo at 06 49 the following day.

Find, in hours and minutes, the time taken for the whole journey.

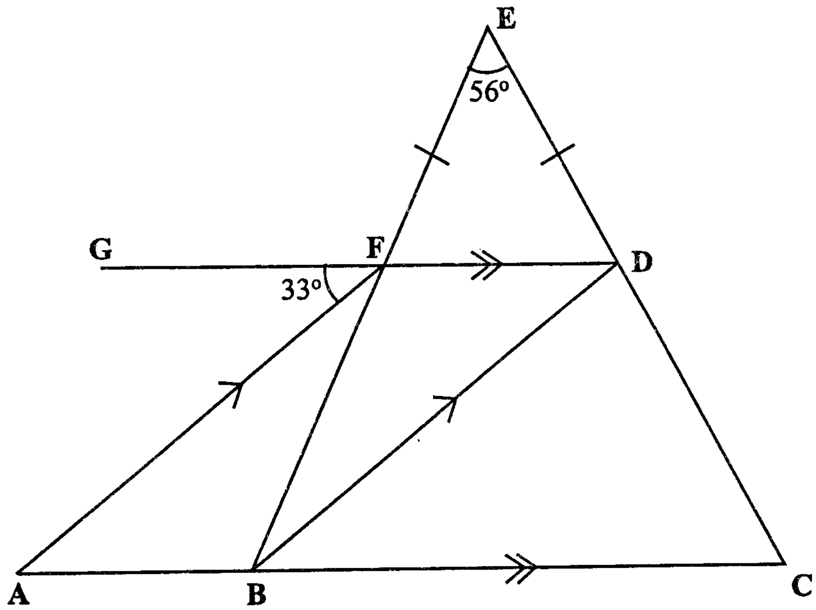
Answer *(a)* ____________________ [1]

(b) ________ h ________ min [2]

[Turn over

For Examiner's Use

15



In the diagram above, ABC, GFD, BFE and CDE are straight lines.
ABC is parallel to GFD and AF is parallel to BD. $D\hat{E}F = 56°$, $A\hat{F}G = 33°$ and $FE = DE$.

Find

(a) $D\hat{F}E$,

(b) $B\hat{C}D$,

(c) $F\hat{B}D$.

Answer *(a)* $D\hat{F}E =$ ______________ [1]

(b) $B\hat{C}D =$ ______________ [1]

(c) $F\hat{B}D =$ ______________ [1]

For Examiner's Use

16 Given that $T = W + Wv^2$,

(a) express W in terms of T and v,

(b) find the value of W when $T = 91$ and $v = 5$.

Answer *(a)* W = ______________________ [2]

(b) W = ______________________ [2]

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17 (a) Given that $f(x) = 3x^2 - x$, find $f(-2)$.

(b) Simplify $3(x + 2y) - 2(x - y)$.

Answer *(a)* $f(-2)$ = ______________________ [2]

(b) ______________________ [2]

18

Mark	5	6	7	8	9
Number of pupils	12	13	14	5	6

The above distribution shows marks obtained by 50 pupils in a class assignment.

Find

(a) the mode,

(b) the median,

(c) the mean mark of the class.

Answer *(a)* ______________________ [1]

(b) ______________________ [1]

(c) ______________________ [2]

For Examiner's Use

19 **(a)** During a drought, a farmer lost 20% of his herd of cattle and was left with 400 only. Calculate his original herd.

(b)

SPECIAL OFFER!!
40% OFF
ON ALL MARKED PRICES

A shop displayed the above advertisement on its windows.

Calculate

(i) the selling price of an article marked \$2 500 in this shop,

(ii) the amount saved by buying the article in this shop.

Answer *(a)* ______________________ [1]

(b) *(i)* \$______________________ [1]

(ii) \$______________________ [2]

For Examiner's Use

20 **(a)** **(i)** Solve the inequality $5 - 2x > 11$.

(ii) Illustrate the solution on the number line in the answer space below.

(b) Solve the equation

$(4t + 3)(t - 3) = 0$.

Answer *(a)* *(i)* ______________________________ [1]

(ii)

-3	-2	-1	0	1	2	3

[1]

(b) t = ____________________ or ____________________ [2]

For Examiner's Use

21 **(a)** A map of a town is drawn to a scale of 1 cm:5 km.

(i) A road on the map is 8 cm long. Calculate the actual length of the road, giving the answer in kilometres.

(ii) The actual area of the town is 150 km^2. Calculate in square centimetres, the area of the town on the map.

(b) Explain what supplementary angles are.

Answer *(a)* *(i)* ____________________ km [1]

(ii) ____________________ cm^2 [2]

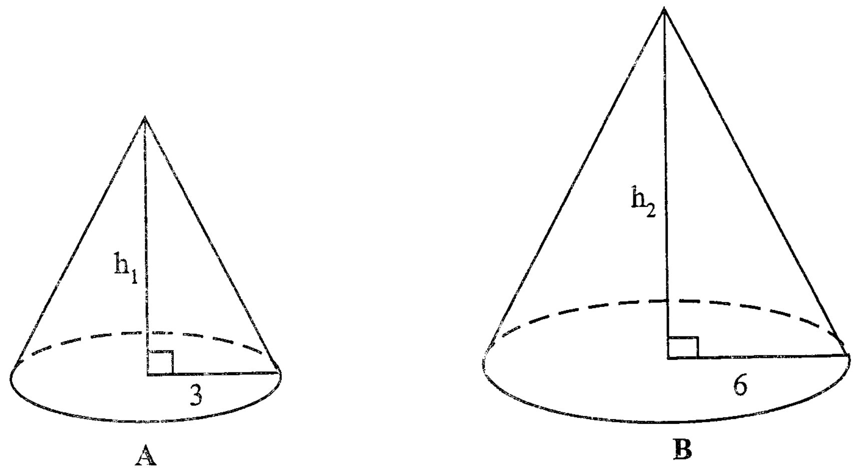
(b) ______________________________

____

______________________________ [1]

For Examiner's Use

22 (a)



Use $\pi = \frac{22}{7}$

The diagram above shows two similar cones A and B. The radius of A is 3 cm and that of B is 6 cm.

(i) Write down the ratio h_1:h_2 in its simplest form.

(ii) Calculate the volume of B, given that the height of A = 7 cm.

(b) Calculate the curved surface area of a cone of radius $10\frac{1}{2}$ cm and slant height 17cm.

[Volume of cone $= \frac{1}{3}\pi r^2 h$

Curved surface area of cone $= \pi rl$]

For Examiner's Use

Answer (a) (i) $h_1 : h_2$ .... [1]

(ii) .... cm^2 [2]

(b) .... cm^2 [2]

For Examiner's Use

23 A packet contains 8 red sweets, 5 green sweets and 7 yellow sweets. All the sweets are identical except for colour.

Two sweets are picked from the packet at random, one after the other, without replacement. Giving each answer as a common fraction in its lowest terms, find the probability that

(a) both sweets are red,

(b) the two sweets are of the same colour,

(c) at least one sweet is green.

Answer *(a)* ______________________ [1]

(b) ______________________ [2]

(c) ______________________ [2]

For Examiner's Use

24 Study the pattern below.

Row

1^{st}	$0 \times 8 + 1 = 1 = 1^2$
2^{nd}	$1 \times 8 + 1 = 9 = 3^2$
3^{rd}	$3 \times 8 + 1 = 25 = 5^2$
4^{th}	$6 \times 8 + 1 = 49 = 7^2$
5^{th}	$10 \times 8 + 1 = 81 = 9^2$
6^{th}	________ = __ = __
•	
•	
•	
r^{th}	$p \times 8 + 1 = 441 = 21^2$

(a) Complete the 6^{th} row of the pattern.

(b) Find the values of p and r.

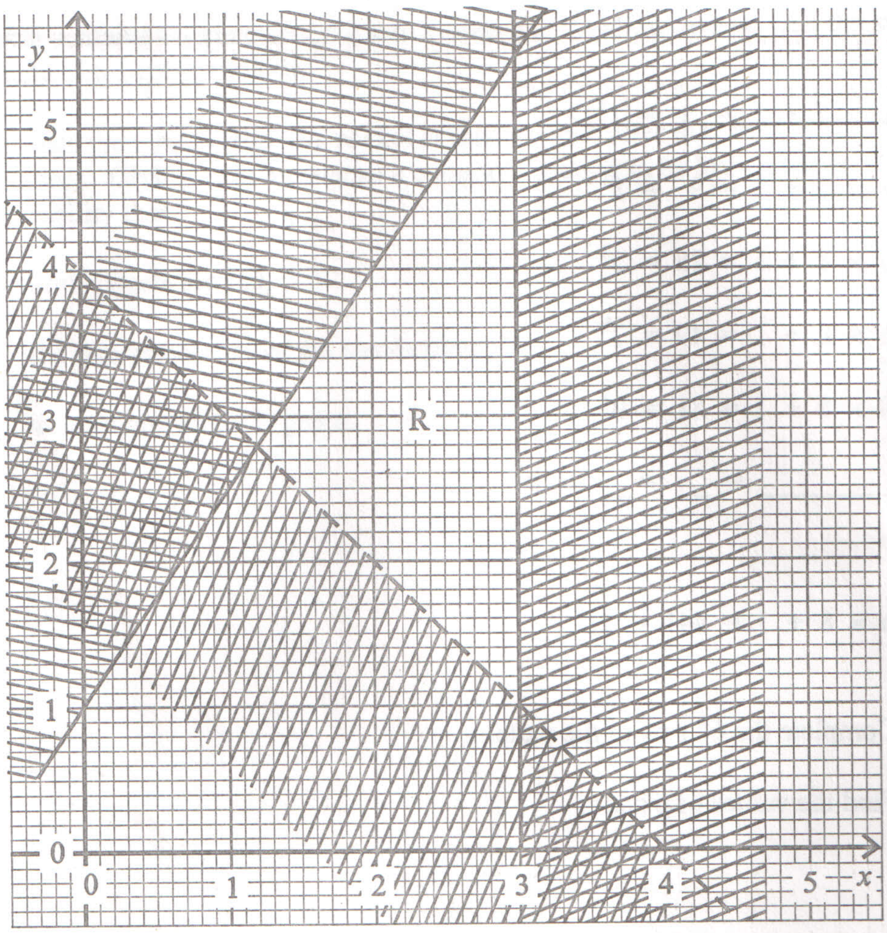
Answer *(a)* on the pattern [3]

(b) p = ______________________ [1]

r = ______________________ [1]

[Turn over

25



Write down the three inequalities which define the **unshaded region** labelled **R.**

For Examiner's Use

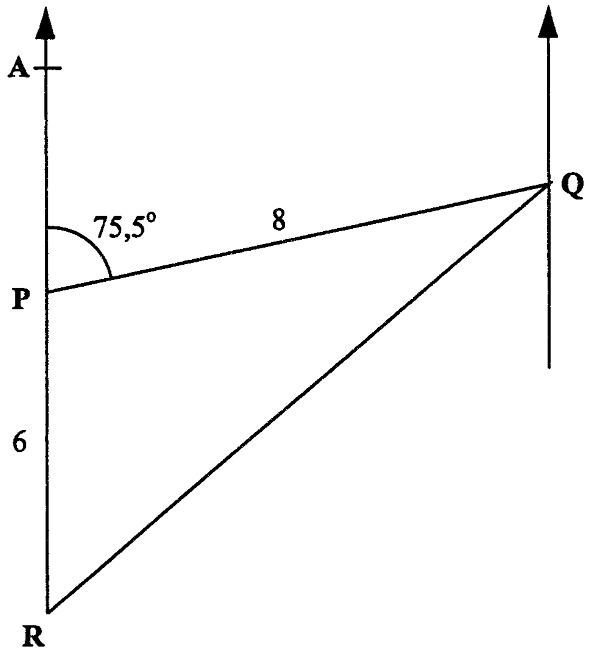
Answer ______________________________

____

______________________________ [5]

For Examiner's Use

26



In the diagram $A\hat{P}Q = 75{,}5°$, $PR = 6$ km, $PQ = 8$ km and APR is a straight line.

(a) Find the three-figure bearing of P from Q to the nearest degree.

(b) Using as much of the information given below as is necessary, calculate

(i) the area of triangle PQR,

(ii) the shortest distance of Q from AR.

[$\sin 75{,}5° = 0{,}97$; $\cos 75{,}5° = 0{,}25$; $\tan 75{,}5° = 3{,}87$]

For Examiner's Use

Answer (a) ____________________ [1]

(b) (i) Area of ΔPQR = __________ km^2 [3]

(ii) ____________________ km [2]

[Turn over

For Examiner's Use

27 **(a)** Simplify $\frac{3}{4}\log 16$.

(b) Evaluate **(i)** $\frac{\log(\frac{1}{5})}{\log 25}$,

(ii) $\frac{\log_2 16 + \log_2 8}{\log_2 8 - \log_2 4}$.

Answer *(a)* ____________________ [1]

(b) *(i)* ____________________ [2]

(ii) ____________________ [2]